

A PLAIN-LANGUAGE GUIDE

Exposure Assessment

How epidemiologists turn contact with an agent into a number, a category, or a yes/no — and why nearly every step of that process can introduce error.

Based on “Exposure Assessment” by Sylvaine Cordier, Charline Warembourg, and Patricia A. Stewart — Handbook of Epidemiology

ROADMAP

Seven Stops

- 01** What exposure assessment means, the forms exposure can take, and how the right metric is chosen
- 02** Measured and modeled data — biological samples, environmental monitoring, models, and personal sensors
- 03** Indirect data — designing questionnaires, keeping diaries, and using existing records
- 04** Building an individual exposure estimate, and the challenge of reconstructing exposure retrospectively
- 05** Studying multiple, correlated exposures — from classic mixture strategies to the exposome
- 06** The types, sources, and quantification of measurement error — and how to correct for it
- 07** Bringing it all back together

FOUNDATIONS

Exposure Is Contact; Assessment Is Classification

Exposure is contact of an individual with an agent through any medium or environment. An agent need not be harmful — exercise or fiber in the diet count too.

Exposure assessment aims to identify whether a person is exposed or not (a dichotomous classification) and, often if exposed, to develop a ranking of subjects by exposure level — in continuous units or ordinal categories — for use in analyzing a disease risk.

A prerequisite for any epidemiological study: there must be **variability** in exposure to the agent within the population being studied — if everyone shares the same exposure, there is nothing left to compare.

FOUNDATIONS

One Word, Many Forms

This chapter uses the single term “exposure” for all of the following, rather than separating external agents from internal ones:

In the external environment

A chemical, biological, or physical agent — cadmium, endotoxin, ionizing radiation

Societal or contextual

A feature of the surrounding context, such as the existence of a support system

Of the individual

A characteristic (weight, physical activity) or a perception (lack of control at work)

Already inside the body

A biological agent (herpes virus), a metabolite (1-hydroxypyrene), a pathway marker (DNA-PAH adducts), or a polymorphism (NAT wild type)

FOUNDATIONS

The True Risk Factor, or a Surrogate for It

Ideally, exposure assessment targets the “true” (causal) risk factor. In practice, however, a **surrogate** often has to stand in — because the true risk factor has not been identified, because some exposures are notoriously hard to quantify (time spent in a malarious area, rather than number of infectious mosquito bites), or because a mixture has so many components that no single causative agent can be isolated (IARC has classified nickel refining as carcinogenic without identifying which compound is responsible).

Surrogates such as pack-years or duration of employment remain useful — for identifying factors of variation, establishing presumptive causal associations, monitoring exposure, and narrowing the search for the true risk factor

The trade-off: precisely because it is a surrogate, it is likely subject to more misclassification than the true risk factor itself would be

FOUNDATIONS

Dose Versus Exposure

Exposure usually refers to contact with an agent in the external environment. A full measure should account for physical state (liquid, gas, or solid), route (inhalation, dermal, ingestion), time spent in each setting, and the individual's own susceptibility — exercise, diet, genetics.

All of these variables shape the internal *dose* that actually reaches human tissue or fluid. A biological marker of internal dose can therefore come closer to the truly relevant measure than an external exposure estimate can, in some circumstances.

Convention used throughout: **exposure** describes the agent being estimated for an exposure-response relationship; **dose** describes the level of the true risk factor that reaches the target organ.

FOUNDATIONS

Choosing the Right Metric

The metric should reflect the agent's characteristics, its toxic mechanism, and the disease of interest.

Chronic disease (e.g., cancer)

Thought to result from lifetime exposure — the usual metric is **cumulative** exposure

Acute disease (e.g., asthma)

Thought to be triggered by recent high exposure — the usual metric is **peak** exposure

A biological threshold

Below which detox is not impaired — **average exposure, or peak exposure above the threshold**, may matter more than lifetime cumulative exposure

The biological mechanism is often unknown, so exploring several metrics at once — cumulative, highest, average, time above a threshold — is generally recommended. Choosing a metric that does not fit the disease process risks misclassification and lost statistical power.

FOUNDATIONS

Pack-Years, Latency, and Lagged Exposure

Pack-years (years smoked $\times$ average packs per day) illustrates cumulative exposure: the same total can be reached by high intensity over a short period or lower intensity over a long one. The index does not distinguish between these two roles — duration and intensity — even though they may carry different biological weight.

Latency: carcinogenesis can be initiated at any dose, but the induction period before disease appears is often long — sometimes decades. Because recent exposure may not be biologically relevant to a diagnosis made now, investigators often “lag” the exposure window, excluding the years immediately preceding diagnosis from the metric.

EXPOSURE DATA

Two Kinds of Exposure Data

Direct

Measured or modeled levels — objective data from instruments or predictive models

Indirect

Questionnaire information, diaries, and records of surrogate information

This section covers direct data; the section that follows covers indirect data.

DIRECT DATA

Measurement Data: Generally the Most Accurate

Measurement data are generally considered the most accurate type of exposure data, because they are objective measures of exposure. Examples include:

Radiation, and chemicals in food, air, or water
— in the general environment or the workplace

Physiological measurements, such as blood
pressure or weight

Activity measurements, such as exercise
tracked with an accelerometer

Biological markers in tissue — blood lead, drug
levels, a genetic polymorphism

Measurements may be collected for the purposes of a study, or drawn from existing records.

DIRECT DATA

“Gold Standard” Does Not Mean Error-Free

May not represent the biologically relevant time window — current physical activity, for instance, may not reflect earlier levels

The measurement or analytical method itself may be inaccurate or inappropriate for the question being asked

The number of measurements per person is often small, while exposure variability (in air or water, say) can be large

Historical records were often collected to check regulatory compliance, not to estimate the exposure metric of real interest — and may not have distinguished what current science considers important (older cholesterol measurements did not separate HDL from LDL)

DIRECT DATA, 1 OF 2

Biological Measurements: The Gold Standard, With Caveats

Measurements of exposure (e.g., carbon monoxide in the breath) or of effect (e.g., blood cholinesterase levels) are generally thought to be the gold standard, because they most closely reflect the dose received by the target organ. Limitations remain:

Body concentrations are often more variable than external ones, so many repeated measurements may be needed

May reflect an amount never received by the target organ; correlation with target-organ dose varies by agent, organ, and individual

Few agents have long-term biomarkers; current levels may not reflect long-term exposure

Even with a long half-life, one measurement reflects body burden at a single point in time only

Models can sometimes bridge the gap between a biomarker and the target organ: physiologically based pharmacokinetic (PBPK) models estimate target-organ dose from an agent's absorption, distribution, metabolism, and elimination, while reverse dosimetry works backward from a biomarker concentration — in urine, say — to estimate the original intake.

DIRECT DATA, 2 OF 2

Costly, Sometimes Invasive, Still Valuable

Biological measurements are often costly and invasive. Measuring asbestos in broncho-alveolar lavage specimens is one direct way to evaluate exposure, but it is invasive and impractical — so asbestos exposure assessment instead relies on environmental measurement, questionnaires, and job histories. Because of cost, biological measurements see more use in case-control or cross-sectional studies, or on a cohort subsample, than across an entire cohort. Measurements taken after disease onset may reflect an *effect* of the disease rather than a cause (e.g., serum androgens and prostate cancer).

Their strengths remain real: useful mechanistic insight; good for exposures with low variability over time or that accumulate in the body, or for levels at key windows (in utero, childhood, puberty); and they capture concentrations from **all** sources at once, better estimating total exposure — valuable when individual practices, like hand-washing, affect the internal dose but obscure the primary source.

DIRECT DATA

Environmental Measurements: A Lower Gold Standard

Measurements of the external environment do not capture the internal dose actually received, so they rank below biological measurements — but they are noninvasive, typically cheaper, and analytical methods exist for more agents. Because group-level environmental variability is usually lower than intraindividual variability in the body, a small number of environmental measurements on similarly exposed people can outperform a small number of biological measurements at estimating exposure from one given source.

The catch: this kind of measurement typically reflects only **one** source, when several may contribute (pesticide exposure at work, from nearby farm drift, and from contaminated food and water). Missing other sources can underestimate true exposure and overestimate disease risk.

DIRECT DATA

Modeling: Predicting What Was Never Measured

Modeling means developing a mathematical or statistical relationship — from the agent’s physical and chemical properties and factors known to affect exposure (particle size, wind direction and speed) — to predict exposure levels where none were measured.

Advantages

Needs no comprehensive measurement data, is less costly, requires less observation time, and applies to any population meeting the model’s conditions. Land use regression (LUR) is a classic example, predicting air pollution at unmonitored sites from traffic, altitude, and land use.

Disadvantages

Perceived as a lower standard; must cover all important exposure components, but some may be unrecognized or unmeasurable; applies only to situations resembling those it was built on; requires careful weighting of each variable

DIRECT DATA

Ecological Measurements: One Value for a Whole Group

When data are not available on the actual study subjects, the same exposure level is assigned to individuals thought to be similarly exposed — for example, the same trihalomethane concentration to everyone on a public water supply, even though true exposure varies with water source, amount consumed, and use.

Common in air pollution epidemiology (ambient or modeled levels assigned at the residence) and occupational epidemiology (job-exposure matrices, or JEMs — a cross-tabulation of job, industry, and time linked to an exposure estimate). Ecological data are cheap, but a measured personal exposure can be three to six times larger or smaller than the group's estimate — even among workers thought to be similarly exposed. Using group-level data instead of individual measurement makes the most sense when exposure differs much more **between** groups than **within** one.

DIRECT DATA

Personal Sampling and Wearable Sensors

Personal sampling has long been used in occupational settings — dosimeters, for instance, continuously monitor a worker's exposure to ionizing radiation.

More recently, wearable sensors assess exposure to air pollution, noise, physical activity, and UV radiation in the general population — sometimes coupled to a smartphone's own clock, GPS, and accelerometer. One app, deployed in a panel of 158 adults across five European cities, calculated time spent in different microenvironments, activity level, inhalation rate, and exposures such as green space and inhaled ultrafine particles. This approach works for both adults and children.

DIRECT DATA

Two Practical Issues: Censoring and Quality Control

Censored data: results reported as below the limit of detection (LOD). Simply excluding them biases a summary statistic — a mean or median biased high, a standard deviation biased low. Simple substitution (e.g., LOD/2), maximum likelihood estimation, and Bayesian methods can handle this more properly.

Quality control: when measurements are taken specifically for a study, the sampling strategy should reflect its actual goals (randomly, or randomly within strata; peak versus full-shift), strict quality-control methods should be followed, and exposure conditions documented. When using existing records instead, the collection, analytical, and quality-control methods behind them — and how representative the sample is — should be reviewed.

EXPOSURE DATA

When Exposure Cannot Be Measured Directly

Indirect exposure data is not measured or modeled directly — it is derived from questionnaires, diaries, or records that identify the presence or level of an exposure.

Questionnaires may collect quantifiable data (cigarettes per week) or subjective measures (perceived control at work)

Diaries and records supply indirect data too — milk products consumed, or distance of a residence from a waste site

INDIRECT DATA

Designing a Questionnaire

Questionnaires collect individual characteristics that may determine exposure — place of residence, occupation, time-activity patterns, diet, hobbies.

They are built to collect information in a structured, standardized way — reducing differential questioning between individuals (particularly those whose answers might be shaped by their own exposure or disease status) and ensuring the information needed for exposure assessment is as complete as possible.

A questionnaire is defined by its content, the span of a subject's life it covers, and its format and wording. Common-sense principles should guide construction, with a flow that is clear and open to minimal misinterpretation.

INDIRECT DATA

How a Questionnaire Is Administered

In person, by telephone, by mail, at home, or in a hospital — the setting can affect the response rate. Development, administration, data entry, and curation are all costly and time-consuming.

Computer-assisted personal and telephone interviews (CAPI and CATI) cut data-entry and curation costs versus paper, though they cost more to develop. Logic checks catch errors immediately rather than long after the interview, and skip patterns save interviewers time flipping through pages.

When a questionnaire is not self-administered, professional interviewers — not the scientists who designed it — usually deliver it, so a request for clarification may not get an answer that improves data quality. Interviewer training and probing questions help address this.

INDIRECT DATA

Writing Questions People Can Actually Answer

The question list should include only what respondents can understand — words and concepts alike — and correctly answer.

Match respondents' educational level; a 14-year-old's reading level is generally appropriate for general-population studies

Keep the burden reasonable — one hour is usually the maximum, though retaining real interest may call for much less

Aids help recall — lists of pesticides, product logos and trademarks, pictures of medication bottles

A minimum question set maximizes efficiency; a few extra questions allow cross-checking

“Red herring” questions — a fake product slipped into a list of real ones — are often useful for evaluating how accurately respondents are actually answering.

INDIRECT DATA

Current Exposures Are Easier to Report Than Historical Ones

Respondents report current exposures more easily and accurately than historical ones — but current practices may not reflect past ones, and can actually distort how the past is recalled.

Asking for a single summary value across many years — as chronic-disease etiology often requires — is generally difficult and prone to inaccuracy. Recollection tied to a specific age or life event tends to improve recall over an abstract span of calendar time.

Example: nutrition questionnaires traditionally collected only current diet, yet current diet is not necessarily well correlated with diet 20–30 years earlier, since eating habits change over time.

INDIRECT DATA

Open-Ended Versus Closed-Ended Questions

Open-ended

“What exercise did you do in your twenties?” — can surface exposures the investigator never anticipated, but requires extensive coding, some answers prove useless, respondents at the same true exposure level may still answer differently, and important exposures can still go unrecalled

Closed-ended

“Did you walk? jog? play tennis?” — takes more design time, but respondents are less likely to forget a listed item, generally making data more accurate; an unlisted exposure may go unreported, and response categories for sensitive topics can feel judgmental (0 to >5 partners reads more harshly than 0 to >20)

Open-ended questions often help develop standardized closed formats during pilot studies. Screening questions (a simple yes/no, or a threshold like “for at least one year?”) minimize time spent on detailed follow-up.

INDIRECT DATA

Diaries

Another source of exposure information, used most often for diet and, to a lesser extent, physical activity. The respondent records an amount — red meat consumption, say — at a set frequency, typically daily.

Diaries work best for exposures that occur **frequently** — a rare exposure risks simply being forgotten. Recording time should be minimal (under a minute) and the covered period short (1–2 weeks) to maximize compliance; check boxes ease data entry better than open-ended fields. Diaries, obviously, cannot reflect historical conditions.

INDIRECT DATA

Records: Often the Only Route to the Past

Records are often needed for retrospective assessment: hospitals, clinics, employers, unions, universities, and governments can all be sources. Occupational studies often rely on work histories from employer records; geographic information systems (GIS) and other surrogate records — the amount of corn grown by county, say — are common in ecological studies.

Records may or may not have been accurately collected, and even accurate data can be limited by why they were collected. Example: the US EPA's Toxic Release Inventory only requires reporting above a minimum threshold — so an area with few reported releases may simply have many small emitters below that threshold, not truly low exposure. Its protocol and quality control should be evaluated before use.

THE PROCESS

From Imperfect Data to a Usable Estimate

The process of exposure assessment aims to construct, from data that are usually imperfect and incomplete, a valid and precise individual exposure estimate and classification of subjects.

How straightforward or complicated this is depends on three things: the level of detail and accuracy of the available data (a surrogate may warrant less intensive effort than the true risk factor would); the goal of the study — hypothesis-generating versus hypothesis-testing; and the resources available to the investigator.

THE PROCESS

Some Data Need Little Processing — Others Need a Lot

Closed-ended questionnaire answers — smoking habits, intake of a food group — often need little further processing.

Others must be combined with individual determinants of exposure. Food composition tables, for example, convert food-frequency answers into nutrient totals (total vitamin A across fruits, vegetables, and meats) — and need regular, country-specific updating.

In environmental and occupational epidemiology, estimation is often more complex and retrospective, with no single recognized method. Models may assign weights to determinants; JEMs combine occupational history (jobs, tasks, years) with exposure data by job title, department, industry, and year.

THE PROCESS

Qualitative, Semiquantitative, or Quantitative?

Qualitative

A yes/no or ever/never classification

Semiquantitative / ordinal

Low, medium, high — or a score, with or without quantitative levels attached

Quantitative

With units of measurement

Continuous data give more statistical power to detect an association — but they also create an impression of higher quality than categorical data, which may not be warranted if the underlying exposure data are actually poor. Continuous exposure in a regression typically assumes a linear risk increase per unit; categories fit the observed data more flexibly, without needing that assumption.

A practical principle either way: when feasible, collect exposure at the most detailed level possible, and collapse it into categories later. Detail lost during collection cannot be recovered, while a detailed dataset can still be regrouped with different cut points afterward — to compare results with other studies, or to explore different toxic mechanisms.

THE PROCESS

Categories Feel Safer — They Are Not Necessarily More Accurate

Using categories is often believed to reduce error — it does not. Every individual in a category is assigned the same value, typically the category's median, though it would be rare for everyone in it to truly share that exposure level.

Some individuals really are near the median and are correctly represented. Others sit at the category's low or high edge yet get the same value as everyone else in it. Worse, people on either side of a cut point — the high end of one category and the low end of the next — may be nearly identical to each other in reality, but are assigned to different categories and different values. This within-category variability reduces the ability to detect a true exposure-response relationship, whichever category scheme is chosen.

THE PROCESS

Exposure Groups: Three Kinds of Variability

Exposure groups are subsets of the population viewed as similarly exposed and assigned the same level, often because more detailed individual data is unavailable. A stricter definition (more shared determinants) usually means less variability within a group, but more variability between groups. Groups may be defined at the questionnaire stage, during exposure assessment, or during analysis.

Intraindividual variability

Day-to-day variation within the **same person** — someone averaging two drinks a day may have none some days and four on others

Intragroup variability

Variation among **different people** placed in the same exposure group — some average two drinks a day, others three, others four

Intergroup variability

Variation **across the groups** that have been defined — say, 0, 1–2, 3–4, 5–6, and >6 drinks a day

The goal: narrow, non-overlapping exposure ranges within a group, and as wide a range as possible across groups — the more intragroup variability relative to intergroup, the more likely a true exposure-response relationship is missed.

WORKED EXAMPLE

Coal Dust and Lung Function: Grouping Strategy Matters

A study of coal dust exposure and change in lung function (FEV1, forced expiratory ventilation in one second) tested four exposure-grouping strategies, each evaluated for intragroup and intergroup variability.

Across the four strategies tested

Intragroup variance ranged from 0.18 to 0.35; intergroup variance ranged from 0.20 to 0.23; the FEV1 coefficient ranged from -2.0 to -5.9 mL per mg/m³ of dust — all four strategies found a **decline** in lung function per unit of dust exposure

The best-separated grouping

Lowest intragroup variance (0.18) paired with the highest intergroup variance (0.23) — and it produced the largest estimated decline: -5.9 mL of FEV1 per mg/m³ of dust

A direct empirical confirmation of the previous slide's principle: the narrowest within-group spread combined with the widest between-group spread revealed the clearest dose-response relationship.

THE PROCESS

Reaching Back: Retrospective Exposure Assessment

Investigating a chronic disease with a long induction period compounds the challenge: it demands exposure information reaching further back than instruments were built to capture. Historical information is typically less complete than current information, and when exposure levels differed in earlier periods, grouping individuals is almost always necessary.

One major source is the subject's own memory — questionnaires ask about different past periods (5, 10, 20 years ago, or a certain age), relying on memory and the ability to average exposure over a long span. Occupational epidemiology often instead merges historical measurement data, employer work histories, and workplace conditions into JEMs, or draws on a panel of experts working from a subject's complete work history.

THE PROCESS

When Historical Data Simply Is Not There

Investigators may collect current measurements and assume historical levels were similar

Or compare current to historical exposure determinants, applying weights that reflect the difference, to build systematic, transparent historical estimates

Whichever approach is used, developing retrospective exposure estimates is more challenging — and more prone to inaccuracy — than estimating current exposure. Validation of retrospective assessments is often simply not possible, because the historical comparison data needed to validate them typically do not exist.

MULTIPLE EXPOSURES

Exposures Rarely Arrive Alone

Traditionally, epidemiological analyses have focused on single agents. In reality, exposure to multiple potential risk factors tends to aggregate for an individual, because of:

An actual mixture containing multiple agents — gasoline, for instance

Correlated behaviors — smoking, alcohol, and coffee drinking often travel together

A shared source — fish brings both toxicants like mercury and PCBs, and nutrients like omega-3s

Contextual bundling — urban air pollution, lack of green space, and social deprivation

MULTIPLE EXPOSURES

Not a Classical Confounding Problem

Simultaneous exposure to multiple agents cannot be treated as classical confounding, because the exposures making up a mixture are often highly correlated with each other. Stratified analysis and ordinary multivariate modeling are generally inefficient here, since they handle high collinearity poorly — and interaction among components is not merely a statistical nuisance, but may reflect genuinely different biological mechanisms.

Three research questions typically drive the choice of method: (1) the overall or joint effect of the mixture; (2) which specific agent(s) are responsible for the health effect; (3) which groups of subjects share a similar mixture exposure pattern.

WORKED EXAMPLE

PCBs: One Mixture, Different Mechanisms

PCBs are a persistent industrial compound class of 209 different chemical members (“congeners”). Though banned decades ago, general-population exposure remains, mainly from diet — and because the commercial product was always a mixture of correlated congeners, studying any one congener’s toxicity in isolation is difficult.

Some congeners act like dioxins, binding the aryl hydrocarbon (AhR) receptor and potentially causing cancer — this activity is highest in congeners with no chlorine in the “ortho” position

Other congeners, with chlorine in the ortho position, may instead cause neurological effects

The same mixture, acting through genuinely different biological mechanisms depending on which congeners are involved.

MULTIPLE EXPOSURES

Three Classic Strategies for a Mixture Like PCBs

1. Measure and treat every congener separately. Easy to analyze and reports the disease association for each congener individually, but ignores co-exposure to the others and inflates the false discovery rate across many tests.

2. Select an indicator component. PCB 153, often present at the highest concentration, was frequently used to represent overall exposure — less effort, but the most concentrated congener is not necessarily the most toxic for a given outcome.

3. Create a summary index. Aggregate exposures into one index — a simple sum, or a weighted combination such as a toxic equivalency factor (TEF) scaled to dioxin, producing a total toxic equivalency (TEQ). Efficient, but the weights depend on current toxicological knowledge and may need revision as evidence accumulates.

MULTIPLE EXPOSURES

Modern Statistical Approaches to Mixtures

Method	General Principle	Advantages	Drawbacks
Dimension reduction (e.g., PCA)	Summarize correlated exposures into fewer uncorrelated components	Eliminates multicollinearity; easy to apply; handles many exposures	Components may not be interpretable
Clustering (e.g., HCPC)	Classify participants into groups that share similar exposure patterns	Easy to apply; can handle many exposures	The resulting group(s) may not be interpretable
Variable selection (Lasso, Elastic Net, DSA)	Select a reduced set of variables most associated with the outcome	Handles many exposures and sparsity; supports interaction	A predictive method that may not identify the true causal agent(s)
Mixture models (WQS, BKMR)	Build a weighted index (WQS), or estimate the joint exposure-response function (BKMR)	WQS: easy to apply and interpret; BKMR: supports interaction and nonlinear dose-response	WQS cannot handle too many exposures; BKMR is complex to interpret

Some of these methods can also incorporate prior toxicological knowledge, such as grouping chemicals by family — combining toxicological and statistical knowledge helps overcome the constraints of studying multiple exposures at once.

A BIGGER IDEA

The Exposome

Introduced by Christopher Wild in 2005: the totality of environmental (nongenetic) factors an individual is exposed to **from conception to death**.

Classical epidemiology typically studies a single, or a limited number of, exposures over just one part of a lifetime — childhood, pregnancy, employment. Exposome studies instead aim to consider all potential risk factors across the entire lifetime — a shift especially suited to chronic diseases, given their multifactorial origin and long-term development.

Why pursue this? Exposome studies aim to identify critical windows of vulnerability and the temporal sequence of disease. Weighing multiple exposures together, rather than one at a time, can also improve causal attribution and reduce some of the residual confounding and measurement-error problems that come from studying just one exposure domain.

THE EXPOSOME

What the Exposome Actually Includes

Environmental chemicals and air pollution

Features of the urban environment

Lifestyle and socioeconomic factors

Endogenous “omic” signatures —
metabolomic, transcriptomic, proteomic,
epigenomic, microbiome

“Omic” data can play three different roles in exposome research: as a marker of exposure itself, as an early marker of disease, or as a marker of individual susceptibility. Integrating omics into epidemiological studies can deepen understanding of how disease actually develops.

THE EXPOSOME

Exposome Studies Face Even Bigger Challenges

A very high number of exposures of interest, drawn largely from existing cohort studies; the diversity of exposures leads to **mixed** exposure data (direct and indirect together) and **mixed** assessment methods (individual versus ecological) within the same study — heightening the risk of differential measurement error between exposure estimates.

“Omic” data can be **targeted** (one specific biomarker tied to a hypothesis) or **untargeted** (all available biomarkers, for a fuller but only partial picture).

Statistical challenges are even larger than for ordinary mixtures — closely paralleling genome-wide association studies (GWAS). Three main approaches: (1) exposome-wide association studies (ExWAS), testing each exposure-outcome link separately with multiple-testing correction; (2) variable selection to find which exposures jointly best predict the outcome; (3) clustering and dimension reduction to find subjects who share similar exposome patterns.

BOX 1: A WORKED EXAMPLE

The Human Early-Life Exposome (HELIX) Project

One of the first exposome studies. Started in 2013, it pooled data from 6 European birth cohorts and performed a standardized clinical examination on 1,301 children aged 6–11.

Over 100 exposures

Measured during pregnancy and around the exam, via GIS, biomarkers, and/or questionnaires

Three domains

Urban environment (air, noise, meteorology); chemicals (phthalates, phenols, metals); lifestyle (smoking, diet, activity)

Health outcomes

Pregnancy outcomes, cognition and behavior, anthropometrics, blood pressure, respiratory health

Omic markers

From 44 features for urinary metabolomics to over 386,000 for DNA methylation

BOX 1, CONTINUED

What HELIX Found

Correlations between exposures from **different** domains were weaker than correlations **within** the same domain — suggesting a more limited risk of coexposure confounding for studies focused on just one exposure domain.

ExWAS and DSA (Deletion-Substitution-Addition) algorithms were used, in a simulation study, to estimate single- and multiple-exposure associations; cluster analysis, restricted to a single domain, identified subjects sharing similar patterns.

Worked example — urban exposome and child blood pressure: ExWAS and DSA agreed — higher blood pressure was associated with more air pollution — but green space’s “protective” effect (seen alone in ExWAS) shrank once air pollution was accounted for. Cluster analysis found 2 urban-exposure profiles — one high in pollution, traffic, and noise; another high in pollution and urbanization with less green space — both linked to higher blood pressure.

MEASUREMENT ERRORS

Every Method Carries Some Error

Every type of exposure assessment, in every area of investigation, carries some error.

Appreciating the sources, types, direction, and degree of error allows for a more appropriate interpretation of study results — and can even let an investigator estimate the error's likely effect on the findings themselves. Identifying possible sources of error before a study starts can lead to a better protocol and may allow the error to be reduced outright.

Even when precisely quantifying the error is difficult, its sources, types, direction, and degree should be described as fully as possible, so readers can properly interpret the results.

MEASUREMENT ERRORS

Two Types of Error: Random and Systematic

Random error

Data are randomly distributed around the true mean. Described in terms of **precision** — the reproducibility, or reliability, of a measure.

Systematic error (bias)

The overall mean is erroneously high or low. Described in terms of **validity** — the distance between what was measured and the true value of the target variable.

Both types of error matter. To be useful, an instrument must be both **reproducible** (a similar response under similar conditions) and **valid** (it actually measures what it is supposed to, and identifies the true quantity present).

MEASUREMENT ERRORS

Misclassification Error

When measurement error affects how exposure levels are assigned to study subjects, the result is **misclassification**. If a subject with medium fruit intake is assigned to the high-intake category instead, that subject is misclassified.

Historically, measurement error has been associated more with categorical assessments than with quantitative ones — probably simply because quantitative assessment has been less common in the past.

Misclassification of **confounders** is also a real problem: it reduces the degree to which confounding can be controlled. Incomplete smoking histories, for example, can leave residual confounding when studying other risk factors correlated with smoking — though usually a smaller concern than misclassifying the exposure of main interest.

MEASUREMENT ERRORS

Differential Versus Nondifferential Misclassification

Misclassification is characterized by whether it affects the diseased and nondiseased comparison groups similarly or differently.

Differential

A different amount of error for the diseased versus the nondiseased (or between exposure categories). Can bias the association toward **or** away from the null — in the worst case, producing a spuriously significant or nonsignificant result.

Nondifferential

The same error regardless of disease status. Usually biases relative-risk estimates **toward** the null — though rare, documented exceptions under extreme conditions can bias away from it.

Both types of misclassification can lead to incorrect conclusions.

MEASUREMENT ERRORS

Where Measurement Error Comes From

Faulty design of the instrument itself

Errors or omissions in the protocol for using it

Poor execution of the protocol during collection

Subject characteristics —
poor memory, day-to-day
biological variability

Errors during data entry and analysis

These sources should all be carefully evaluated before relying on any measurement instrument. Direct methods (air or biological sampling) can often have their error reduced through calibration and standard quality-control procedures.

MEASUREMENT ERRORS

Error in Questionnaires, Diaries, and Records

Systematic differences can come from incorrect phrasing of a question that leads everyone to misinterpret it the same way, or from inappropriate or misleading response categories.

Random sources come from poor phrasing that different respondents interpret differently, or from a lack of interest or frustration on the respondent's part.

Pilot testing under realistic conditions — on people like the actual respondents, administered the way it will really be administered — reduces bias and increases precision, since investigators often interpret their own questions very differently than respondents do.

Reporting events from many years ago, or that require mental calculation (like an average seasonal food amount), is more prone to random error than reporting recent or simple-to-recall events.

MEASUREMENT ERRORS

Who Answers — and What They’re Asked — Matters

Sensitive topics

Can produce systematic error — subjects may answer “don’t know” more often when the true number (e.g., of sexual partners) exceeds what feels socially acceptable to report

Proxy respondents

Sometimes necessary when a subject is too sick to interview — proxies generally know less about workplace or lifestyle details than the subject would, an asymmetry when compared against healthy controls answering for themselves

MEASUREMENT ERRORS

One Measurement, High Variability: A Recipe for Bias

A single measurement of an exposure with high day-to-day (intraindividual) variability risks serious misclassification, since one day's reading may look very different from a person's true average. Generally, having only one such measurement increases nondifferential misclassification, biasing results toward the null.

Worked example — bisphenol-A: prenatal exposure to this highly variable chemical is best assessed with repeated urine samples across pregnancy. Using just **one** sample instead of repeated ones was estimated to cause an attenuation bias of **69%** — the true effect underestimated by more than two-thirds from misclassification alone.

MEASUREMENT ERRORS

Thin Occupational Data, Diluted Job-Exposure Matrices

Investigators often restrict workplace information to just jobs, industries, and dates, then apply a JEM to assign exposure from this limited information.

Workers in a job are typically called **unexposed** if the exposure occurs in only a small proportion of that job's workers — misclassifying some who are, in reality, exposed. Symmetrically, a job with a high probability of exposure is called **exposed** overall, even if a small share of its workers are not. The impact depends on the relative size of the exposed versus nonexposed groups.

Detailed task and work-condition data collected at the individual level gives a materially better assessment — this is really a problem of limited data, related to selection bias in general, rather than something unique to exposure variables.

MEASUREMENT ERRORS

Recall Bias and Interviewer Bias

Recall bias

Diseased subjects may reflect on, and take more care providing, their past exposures than nondiseased subjects — producing differently accurate reporting on each side

Interviewer bias

When a human intermediary is aware of — or believes they can tell — a subject's disease status, a different attitude toward diseased versus nondiseased subjects can creep in

Face-to-face interviews carry the most risk, particularly for sensitive questions. Self-administered questionnaires are believed less vulnerable, though the questionnaire's appearance, its cover letter, and the research group can still matter. Computer-assisted telephone interviewing (CATI) has become a popular middle ground, often with high response and little missing data.

QUANTIFYING ERROR, 1 OF 2

Reproducibility Studies

Evaluated by comparing the **same** instrument on the same people over time, or **different** instruments assessing the same construct on the same people at the same time.

A self-administered lifetime physical activity questionnaire asked subjects to recall activity at four ages, twice, a year apart. Intraclass correlations ranged 0.78–0.87, with 0.83 overall for lifetime activity.

A new food-frequency questionnaire was compared against two established dietary questionnaires and telephone 24-hour recalls. Despite design differences, correlations for total energy and 26 nutrients were very similar.

Evidence that carefully designed self-administered food-frequency questionnaires can give reasonably reproducible measures of current nutrient intake.

QUANTIFYING ERROR, 2 OF 2

Validation Studies

Ideally, an instrument is compared to a true gold standard under the conditions it will actually be used in — choosing that gold standard is itself a critical issue.

Biochemical indicators offer a strong, independent comparison point, since their errors are unlikely to correlate with a questionnaire's or an environmental measurement's errors — but they are only useful for what they actually measure, and the body burden must not be affected by the disease itself.

A true gold standard often does not exist, especially for retrospective exposure. Partial validation is sometimes possible — comparing reported jobs to employer records, or reported smoking to medical records.

Validity is also often judged by an instrument's ability to predict disease risk — somewhat circular, since the epidemiological outcome is what is being used to test it. Still, a genuinely good instrument should out-predict a poor one, especially where the true exposure-response relationship is already well established.

CORRECTING FOR ERROR

Validation Studies as a Correction Tool

If a systematic difference between measured and true exposure is known, applying a correction factor can reduce or minimize its effect on the exposure estimate.

Internal validation studies

Everyone in the study gets the imperfect measurement; a more accurate but costlier measurement is also collected on a small, randomly selected subset of cases and controls. Methods like regression calibration then infer a correction by fitting measurement-error models within that subsample.

External validation studies

Serve the same purpose, but draw the more accurate reference measurements from a separate study population instead of an internal subsample.

CORRECTING FOR ERROR

Uncertainty and Sensitivity Analysis

Rather than one fixed correction, investigators can identify a plausible uncertainty **range** around each recorded value. Example: if “How many glasses of wine do you drink?” used categories of <1/day, 1–3/day, 4–5/day, and >5/day, the true values behind each could plausibly range as widely as 0–2, 0–4, 2–6, and 3–10.

Monte Carlo or other simulations show how that uncertainty could shift the resulting disease-risk estimate. Useful, but less than fully satisfactory — as Chatterjee and Wacholder put it: “The best way to reduce bias from measurement error is to improve tools for measuring exposures, including biological markers, environmental samples, and questionnaires.”

WRAPPING UP

The Question Underneath Every Study

Exposure assessment sits underneath every epidemiological study — before a study can find, or fail to find, a real disease association, its exposure of interest has to be classified accurately enough to detect that association in the first place.

Whether the data come from a costly biomarker, a single environmental measurement, a carefully piloted questionnaire, or an employer's old records, the same question recurs: how well does what was actually measured represent the true exposure the researcher cares about?

WRAPPING UP

Two Challenges Reshaping the Field

Multiple, correlated exposures

Real-world exposures rarely occur one at a time. Modern methods — dimension reduction, clustering, variable selection, mixture models — are increasingly needed to disentangle overlapping, collinear exposures, and to treat interaction between them as a real biological phenomenon, not just statistical noise.

Cumulative, lifetime exposure

Rather than a single agent at a single life stage, the exposome concept pushes epidemiology to consider the totality of nongenetic exposures across an entire lifetime, including critical vulnerable windows — at the cost of far greater measurement and statistical complexity.

BOTTOM LINE

Match the Method to the Disease — and Own the Error

Accurate exposure assessment means matching the metric, the data source, and the classification method to the disease process under study and the resources actually available — cumulative or peak exposure, biological or environmental measurement, quantitative or categorical classification, one agent or a whole mixture.

No single choice is right for every study. What stays constant is the need for honesty about the measurement error that remains — its likely sources, types, and direction — so a study's results can be properly and fairly interpreted.

The demand for accurate, validated, cost-effective exposure assessment keeps growing as epidemiology takes on lower exposure levels, more complex mixtures, and disease that unfolds across an entire lifetime.

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